

Setup

$$f(x,y) = 2x^2 + 3y^2 \quad \nabla f = (4x, 6y), H = \begin{pmatrix} 4 & 0 \\ 0 & 6 \end{pmatrix}$$

$$x^0 = (5, 1), v^0 = (0, 0), \alpha = 0.1, \beta = 0.6$$

$$v \leftarrow \beta v + \nabla f(x), \text{ then } x \leftarrow x - \alpha v$$

Step 1

$$\nabla f(x^0) = (20, 6) \quad v^1 = 0.6 \cdot (0, 0) + (20, 6) = (20, 6)$$

$$x^1 = (5, 1) - 0.1 \cdot (20, 6) = (3, 0.4)$$

Step 2

$$\nabla f(x^1) = (12, 2.4) \quad v^2 = 0.6 (20, 6) + (12, 2.4) = (24, 6)$$

$$x^2 = (3, 0.4) - 0.1 \cdot (24, 6) = (0.6, -0.2)$$

K	x^k	$\nabla f(x^k)$	v^k	$f(x^k)$
0	(5, 1)	(20, 6)	(0, 0)	53
1	(3, 0.4)	(12, 2.4)	(20, 6)	18.48
2	(0.6, -0.2)	(2.4, -1.2)	(24, 6)	0.84

Since $v^0 = 0$; $x^1 = (3, 0.4)$ Step 1 is identical
 Step 2 $x^2 = (3, 0.4) - 0.1(12, 2.4) = (1.8, 0.16)$, $f = 6.5568$.

After two steps momentum reaches $f = 0.84$ against
 GP's 6.5588, so it moves along x faster

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